

Visualization of Hydraulic Reversal at Low Head Dams through Use of an Acrylic Dam Simulator

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The number of low-head dams in the U.S. (frequently called weirs outside of the U.S.) is difficult to establish. Various sources indicate that there may be as many as 2.4 million, with the Army Corps of Engineers listing more than 76,000 dams in its National Inventory of Dams at least 6 feet in height.¹ In a survey for the National Association of State Boating Law Administrators (NASBLA) 28 states indicated having a range of 2 to 298 low head dams. Several mid-western states that experience low-head dam deaths each year do not have data on the number of dams.² Many of these dams no longer serve their intended purpose, but continue to degrade the natural river environment and are a hazard to boaters, anglers and other people using rivers for recreational purposes. Often these dams are the only features interrupting a normally slow and quiet moving stream, making boaters believe that the water below the dam poses no hazard. However, when water flows over the face of the dam, gravity pulls it towards the bottom of the river causing a depression that gets filled by water flowing back upstream. A person thus is caught by being pushed down and pulled back when the water recirculates. If the dam stretches across the whole width of the river, escape is only possible by diving towards the bottom to catch the outflow (Figure ■1).

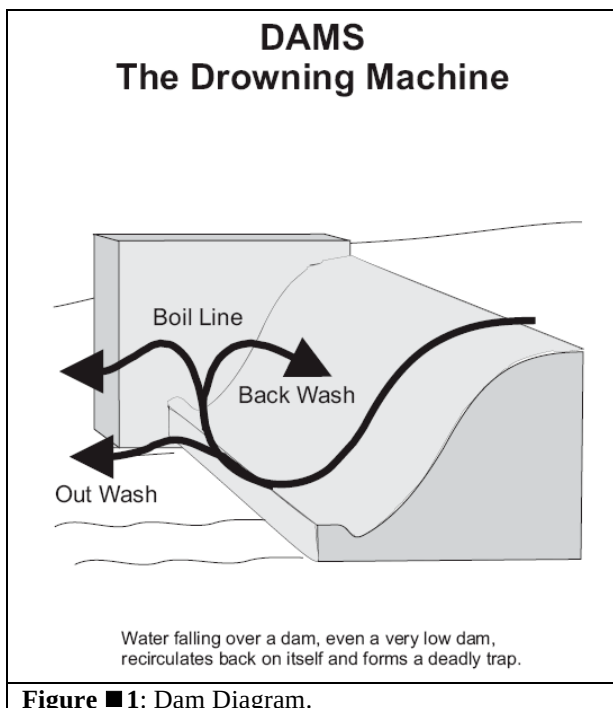


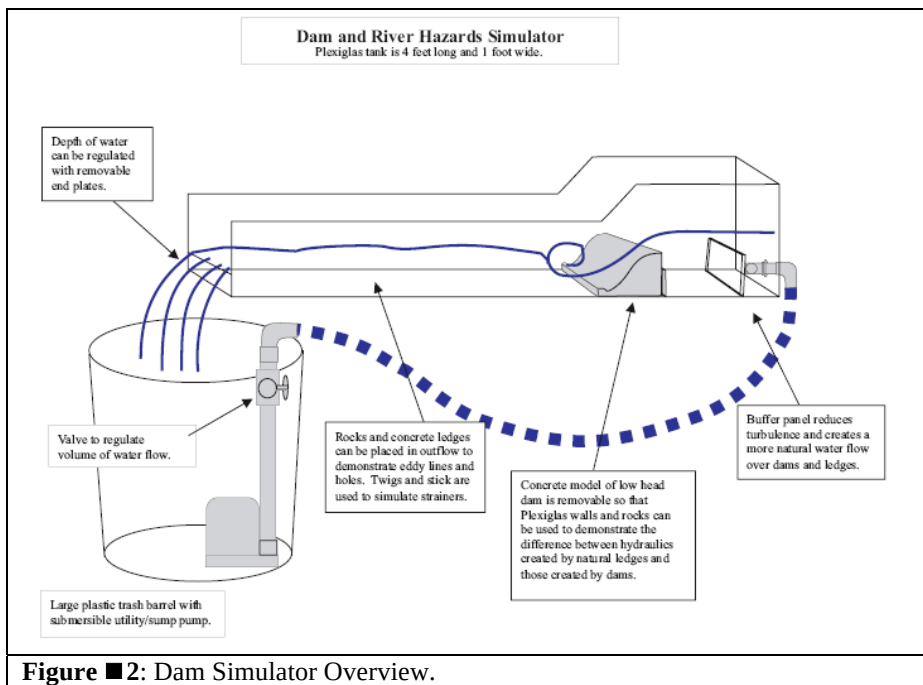
Figure ■1: Dam Diagram.

In the State of Illinois, where the author resides, each year people drown because they are unaware of the hazards of the water flow reversals caused by low-head dams. In 2006, a kayaker went over a local dam. He was witnessed by two young men who ran to his

assistance. All three drowned. At least 13 other people have drowned on this dam since its construction in 1960.³ Ten weeks later, a four-year old boy fell into the water below a well marked dam on another river as his family was fishing. His mother and two other relatives drowned while trying to rescue the child.⁴ As a result of the first tragic accident, the Illinois Paddling Council, of which the author is a former president and current director, released an eight minute video explaining the dangers of low-head dams (www.illinoispaddling.org– click on the left on DAM SAFETY). This video also includes footage of an acrylic dam simulator which clearly visualizes the recirculation that occurs at the downstream side of a dam.

Components of the Dam Simulator

The dam simulator was designed by Susan Sherrod and built by Joel Neuman (**Figure ■2**). It is made from acrylic sheet and standard PVC plumbing pipe components. It is four feet in length and one foot wide. A sump pump submerged in a 40 gallon plastic barrel recirculates the water through a plastic hose. A valve at the dam intake allows for variable flows.



Three different size clear acrylic panels inserted at the overflow end of the simulator make it possible to raise or lower the water level according to what is to be demonstrated. (**Photo ■1**).



Photo ■ 1: Dam setup.

Dam replicas (modeled after typical U.S. Army Corps of Engineering dams with their upturned lip) are made from either concrete or acrylic. Another variable is a natural rock ledge which many times can cause similar hydraulics at the downstream face as do man-made dams. The demonstration includes use of small plastic doll-house people (**Photo ■ 2**). Additional items are cut-out and painted wooden people, small barrels, tires, and similar debris that is often seen recirculating below a dam.



Photo ■ 2: Ken and Barbie in canoe.

The simulator rests on a foldable stand with adjustable feet to ensure that it rests firmly on the ground. All components pack into the barrel and a plastic bin when not in use (**Photo ■ 3**).



Photo ■3: Simulator not in use.

Demonstrations with the Dam Simulator

The dam simulator can demonstrate a variety of natural river features and is most importantly used to demonstrate the dangers of dams:

- Hydraulic reversal that occurs at the downstream face of a low head dam; because the sides of the dam are clear, even the water bubbles that form are visible as they recirculate towards the face of the dam (**Photo ■4**).



Photo ■4: Recirculating person.

- The trained eye also can discern the boil line. Where water starts flowing downstream again, this boil line can be made very visible at high water speed and low water flows.
- A view from above the dam (looking downstream) makes it possible to discern the horizon line.
- Foot entrapment (do not stand up in swift current if water is at your knee) (**Photo ■5**).



Photo ■5: Foot entrapment.

- Strainers (trees and logs that have fallen into the river) (**Photo ■6**).



Photo ■6: Strainer.

- Undercut rocks (especially dangerous in certain areas of the country)
- Other river features as eddies, eddy lines, downstream 'v's', chutes, and more
- Varying the water level to completely submerge the dam, then gradually lowering the water level to demonstrate the appearance of the hydraulic reversal
- Varying the water speed to demonstrate its effect below the dam
- Adding rocks of varying sizes at the face of the dam demonstrates how the dangerous hydraulic reversal can be mediated (**Photo ■7**).



Photo ■6: Remediation.

Comments from the Public and Authorities

The dam simulator has received significant interest from many people and professional organizations, even those very knowledgeable about the dangers of dams. As a result of the accidents mentioned above and the release of the video, the State of Illinois has passed a law to clearly mark all dams owned by the state⁵ (**Figure ■3**).



Figure ■3: Sign.

However, as one comment received by the author noted: “...*your video will have far more impact on the safety of the users of our waterways than anything the legislature can mandate*”.

Conclusion

It is the author's hope that publicizing the availability of the video and the education it conveys will make more people aware of the hazards of low head dams and avoid becoming unintended drowning victims.

References

1. Personal Communication, Serena McClain, American Rivers, Washington, D.C.
2. Personal communication, Bruce Tschantz, PE, Professor Emeritus, University of Tennessee, Civil & Environmental Engineering Department
3. Chicago Tribune, May 29, 2006
4. Chicago Tribune, August 14, 2006
5. Chicago Tribune, May 17, 2007

Credits

- Dam Safety Video Producer, (actual dam footage, compiler, editor, voice over): Thomas Lindblade, (Photo 7) retired Experiential Educator, and Instructor/Trainer for the American Canoe Association. He is the videographer for numerous paddling related videos which can be accessed at <http://www.hauntedbywaters.com>
- Dam simulator and video made possible through two grants by the Baxter Allegiance Foundation
- Dam Simulator Design and Engineering: Susan Sherrod
- Dam Builder: Joel Neuman, B&N Products, Palatine, Illinois, USA, boatsmith@speednetusa.com
- Video footage of the Dam Simulator: Marge Cline, American Canoe Association, Canoe and Kayak Instructor Trainer